

A friendly tour • no coding knowledge needed



Meet MedFlow

A giant digital hospital "brain" that safely collects patient information, spots danger early, and shows it to the right people — built entirely with **pretend (synthetic) patients**.

THE BIG IDEA IN ONE BREATH

What does MedFlow actually do?

Imagine every machine in a hospital — heart monitors, lab computers, X-ray scanners, smartwatches — all speaking **different languages** and scribbling notes on **different forms**. MedFlow is the super-organized assistant that **translates everything into one language, files it neatly, notices when a patient is getting sick, and locks the records away safely** so only the right doctor can see them.

THINK OF IT LIKE...

A very smart, very careful hospital receptionist 🧑



A great receptionist does four things at once:

- 🗣️ **Understands everyone** — no matter what device or form they bring.
- 📁 **Files it perfectly** — so anything can be found again in seconds.
- 🚑 **Raises a hand early** — "this patient looks like they're heading for trouble."
- 🔒 **Guards privacy fiercely** — locks records and writes down *exactly* who looked at what.

MedFlow does all four — but for thousands of patients, instantly, around the clock.

WATCH IT WORK

The journey of a single heartbeat 💖

Follow one reading from a patient's wrist all the way to a doctor's screen. The glowing dot is your data travelling:



Smartwatch

"Heart rate is 128 & climbing"



Front door

MedFlow catches the reading



The library

Filed away safely



Smart helper

"This looks like early sepsis!"



Doctor's screen

Instant alert pops up



⌚ This whole trip happens in **seconds** — fast enough to help in a real emergency.

The 5 main parts (in plain English)

Each part has a job. Hover the cards — the blue note explains the "coder words" simply.



The Front Doors

Special doorways that understand hospital machines — heart monitors, labs, X-ray scanners, watches — and turn their messages into **one shared language**.

Coder words: "FHIR, HL7, DICOM" are just **standard formats** hospitals agree to use — like everyone emailing in the same file type.



The 3-Floor Library

A giant storage building. **Ground floor** = raw papers as they arrive. **Middle floor** = cleaned & tidy. **Top floor** = organized for big research questions.

Coder words: a "data lakehouse" with **bronze** → **silver** → **gold** layers. Each floor up is cleaner and more useful.



The Smart Helpers

Trained assistants that have "seen" many patients and learned the early warning signs — for sepsis, for re-admission risk, and for reading chest X-rays.

Coder words: "machine learning models." They **learn patterns from examples**, then make predictions on new patients.



The Screens & Apps

What people actually see: a **doctor dashboard**, a **patient website**, and a **phone app** — each showing the right info to the right person.

Coder words: "front-end apps" built with React / Next.js — the friendly buttons-and-charts layer humans click on.



The Security Guard

Locks sensitive details, checks **who is allowed** to see each patient, and keeps a **tamper-proof logbook** of every single peek at a record.

Coder words: encryption + "audit log." The logbook is **hash-chained** — if anyone edits an old entry, the chain visibly breaks.



The Privacy Eraser

Before data is used for research, it scrubs out names, phone numbers and addresses — so no one can tell **who** a record belongs to.

Coder words: "de-identification" (HIPAA Safe Harbor). It removes the 18 details that could identify a real person.

"Show me some actual code!" 🙄

Don't worry — here's a tiny, made-up example. Code is really just a **recipe**: a list of clear instructions a computer follows top to bottom.

```
# RECIPE: decide if a patient might be getting sepsis

def check_sepsis_risk(patient):
    heart_rate = patient.heart_rate # grab the latest number
    temperature = patient.temperature

    if heart_rate > 120 and temperature > 38.5:
        return "⚠️ HIGH RISK – alert the doctor now"
    else:
        return "✅ Looks okay – keep watching"
```

Line 1 A **comment** — a human note the computer ignores. Coders leave these to explain themselves.

def ... "**def**" means "define a recipe." We named this recipe *check_sepsis_risk*.

if / else A **decision**: if the numbers are scary, do one thing; *else* (otherwise) do another.

return The recipe's **answer** — the result it hands back. In MedFlow the real version is far smarter (it learned from thousands of cases), but the idea is exactly this.

A mini-dictionary for non-coders 📖



A "service"

One small program that does **one job well** — like one worker on a team. MedFlow has **14 workers**, each handling its own task and talking to the others.



An "API"

A **menu of requests** one program offers to others. Like ordering at a restaurant: you ask for what's on the menu, and you get it back — without seeing the kitchen.



A "database"

A super-powered **filing cabinet** that can store millions of records and find any of them instantly.



A "container"

A **sealed lunchbox** holding a program plus everything it needs to run — so it works the same on any computer, anywhere.



The "cloud"

Someone else's powerful computers, rented over the internet, so the system can grow to serve **more patients** on demand.



"Real-time"

Things happening **right now**, live — like the sepsis alert popping onto a doctor's screen the instant the data arrives.

BY THE NUMBERS

How big is this thing?

14

mini-programs
(services)

833

files written

3

smart AI helpers

41

moving parts
running together

0

real patients used
(all pretend!)

Why build something this careful?

In healthcare, a mistake or a leak can hurt a real person. So MedFlow was built to prove one thing: that software can be **fast and smart and private and trustworthy** at the same time — the gold standard for any serious medical system.

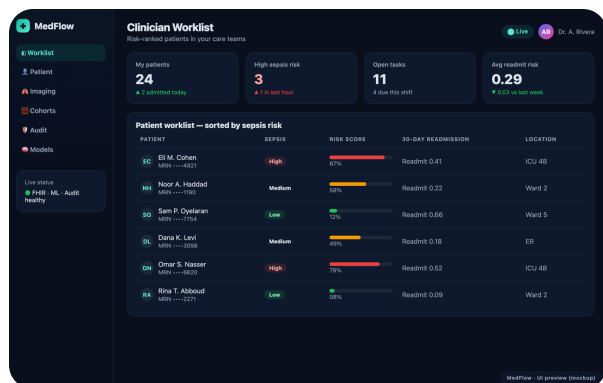
One important promise 🍷 — Every "patient" in MedFlow is **completely made up** by a generator called Synthea. No real person's information is ever used. It's like a **flight simulator**: realistic enough to learn and prove skills, but nobody is actually on board.

SEE IT IN ACTION

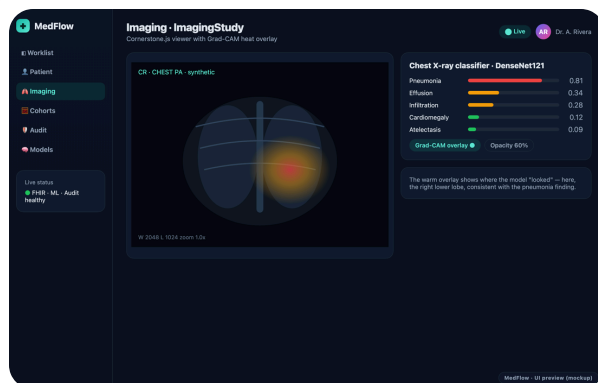
What the screens look like

A quick tour of the real screens people would use. *(These are polished **previews / mockups** — they show the design and the kind of information shown, not live patient data.)*

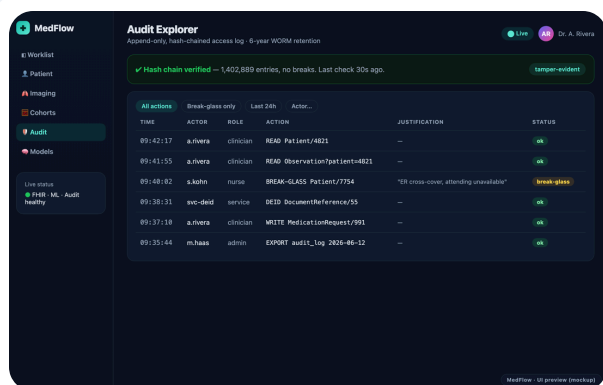




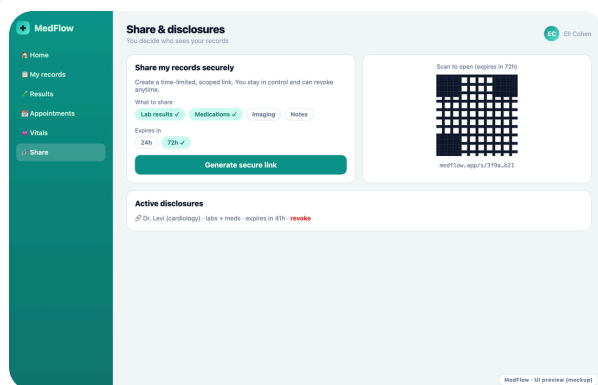
Doctor's worklist — patients sorted by how worried we should be.



X-ray with AI "heat map" — the glow shows where the computer looked.



Tamper-proof logbook — every peek at a record is recorded forever.



Patient stays in control — share records with a time-limited link.



MedFlow — Scalable Healthcare Data Platform

A portfolio project showing how to build safe, smart software for hospitals.



Healthcare data



AI / ML



Privacy & security



Cloud-native



Big data



Web & mobile

Built by **Yousef Hasan Hamda** · [LinkedIn](#) · For the full technical details, see the project's [README](#).